THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Who is called the Father of Genetics? 1

(a) Charles Darwin **(b)** Robert Hooke
(c) Louis Pasteur **(d)** Gregor Johann Mendel

2. In a dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) seeds, the F1 seeds are 1

(a) wrinkled yellow **(b)** round green
(c) all round yellow **(d)** all wrinkled green

3. In human beings, the sex of the child is determined by 1

(a) the egg alone
(b) the type of sperm that fertilises the egg
(c) the mother's diet
(d) environmental temperature

4. Mendel chose the garden pea for his experiments because 1

(a) it reproduces only asexually
(b) it grows only in cold regions
(c) it has many contrasting characters and is easy to grow
(d) it has no flowers

5. The F2 phenotypic ratio in the dihybrid cross is 1

(a) 1 : 1 : 1 : 1 **(b)** 9 : 3 : 3 : 1 **(c)** 3 : 1 **(d)** 1 : 2 : 1

6. A normal human female has the sex chromosomes 1

(a) YY **(b)** XY **(c)** XX **(d)** XO

7. The transmission of characters from parents to offspring is called 1

(a) regeneration **(b)** evolution **(c)** variation **(d)** heredity

8. In the dihybrid F2, the number of round green seeds out of 16 is 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 9

9. A normal human male has the sex chromosomes 1

(a) XX **(b)** YY **(c)** XY **(d)** XXY

10. When pure tall (TT) and pure dwarf (tt) pea plants are crossed, the F1 generation is 1

(a) three tall to one dwarf **(b)** all tall

(c) all dwarf **(d)** half tall and half dwarf

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the difference between heredity and variation? 2

12. What is a monohybrid cross? Give the F2 ratio. 2

13. What is an acquired trait? Give an example. 2

14. Why did Mendel choose pea plants? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A pure tall plant is crossed with a pure dwarf plant. Write the genotype and phenotype of F1 and the F2 ratio. 3

16. Explain why the father, not the mother, determines the sex of the child. 3

17. What is the role of DNA in heredity? 3

OR

How do Mendel's experiments show that traits are inherited independently? 3

18. Differentiate between dominant and recessive traits with the help of a pea example. 3

OR

Explain the term 'F1' and 'F2' generations. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain Mendel's monohybrid cross with tall and dwarf pea plants. Show the F1 and F2 generations with a Punnett square and the ratios. 5

OR

Explain Mendel's experiments and the conclusions drawn from them. 5

20. Describe Mendel's dihybrid cross using round yellow and wrinkled green seeds. Give the phenotypes and ratio in F2. 5

OR

A cross is made between tall plants (Tt) and dwarf plants (tt). Show the result with a Punnett square and give genotypic and phenotypic ratios. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND